

Differential Geometry on Triangle Meshes

One matrix — the cotangent Laplacian — from curvature to spectra to flow, in Wolfram Language

Abstract. Smooth surfaces carry curvature, a Laplace operator, a spectrum and a notion of flow. A triangle mesh has none of these a priori — yet *discrete differential geometry* (DDG) builds operators on meshes that reproduce the theorems of the smooth theory exactly. This notebook develops the core of DDG in pure Wolfram Language and validates every quantity against a smooth reference: angle-defect Gaussian curvature (checked by Gauss–Bonnet), the cotangent Laplace–Beltrami operator, mean curvature, the Laplacian spectrum (which recovers the sphere's eigenvalues $l(l+1)$), and mean-curvature flow. A single matrix — the cotangent Laplacian — drives all of it.

1. Two curvatures from one mesh

A smooth surface has two curvatures at each point: the **Gaussian curvature** K (intrinsic — bending that a flat sheet cannot reproduce) and the **mean curvature** H (extrinsic — how the surface sits in space). On a triangle mesh both become per-vertex numbers.

The **angle defect** concentrates all Gaussian curvature at the vertices: for an interior vertex i with incident triangle angles θ , the integrated Gaussian curvature is the failure of those angles to close up to a full turn,

$$K_i = 2\pi - \sum_j \theta_j$$

Summed over a closed mesh this gives the **discrete Gauss–Bonnet theorem**, an exact identity tying local geometry to global topology (χ is the Euler characteristic):

$$\sum_i K_i = 2\pi\chi$$

Our implementation reproduces it to machine precision: the sphere ($\chi = 2$) totals 4π to a residual of 10^{-12} , and the torus ($\chi = 0$) totals zero. The pointwise curvature $K = (\text{angle defect})/(\text{vertex area})$ matches the analytic torus curvature to a relative L2 error of 5×10^{-4} .

Discrete Gaussian curvature on a torus (angle defect)

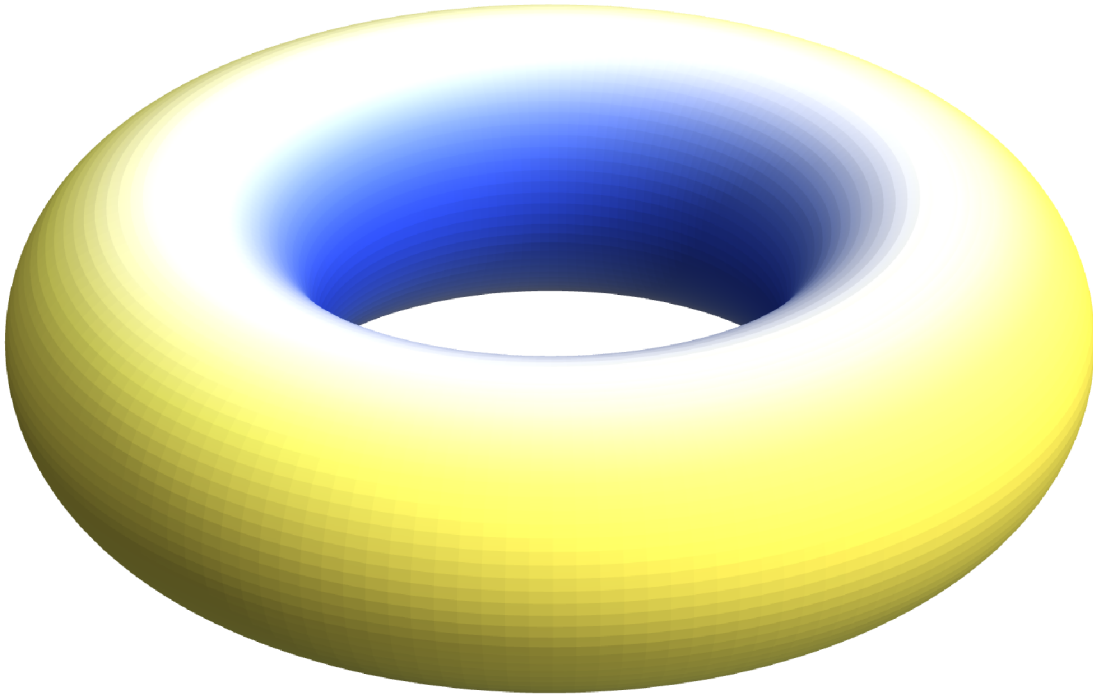


Figure 1. Discrete Gaussian curvature on a torus from the angle defect: positive (warm) on the outer rim, negative (blue) on the inner rim, summing to zero as Gauss–Bonnet demands.

The **mean curvature** comes from the cotangent Laplacian of §2 applied to the vertex positions (the mean-curvature normal of Meyer et al. 2003); it matches the smooth torus mean curvature to 3×10^{-4} and the unit sphere's $H = 1$ to 5×10^{-3} .

Discrete mean curvature on a torus (cotan Laplacian)

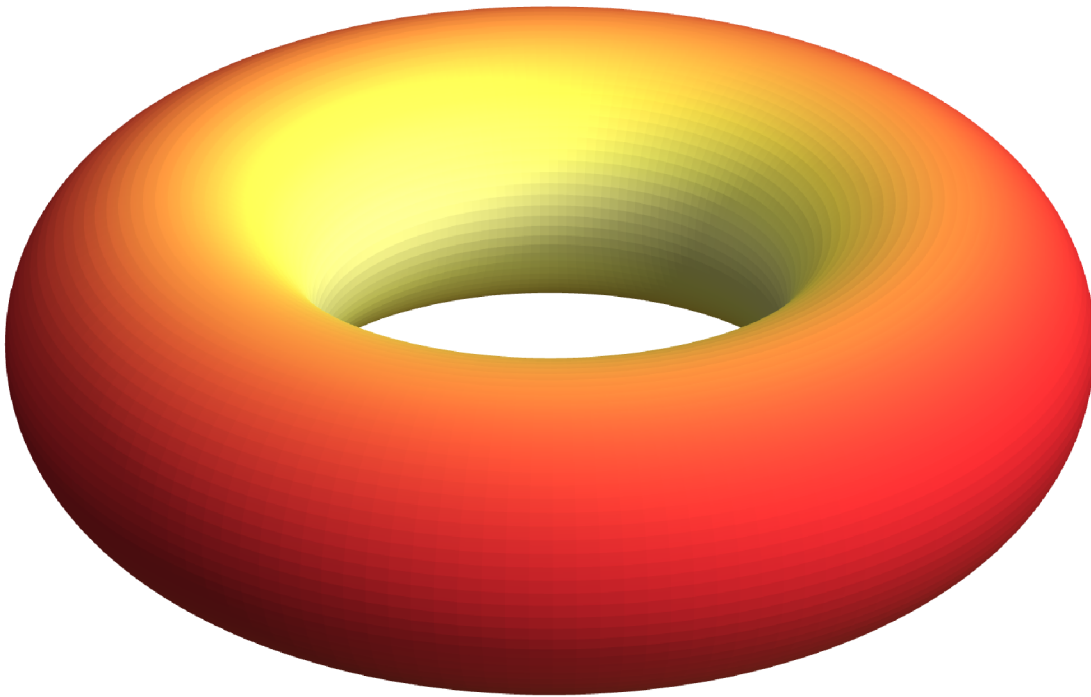


Figure 2. Discrete mean curvature on the torus, computed from the cotangent Laplacian of the coordinate functions.

2. The cotangent Laplace–Beltrami operator

The workhorse of DDG is the **cotangent Laplacian**. For a function f sampled at vertices,

$$L f_i = \frac{1}{2} \sum_j (\cot(\alpha_{ij}) + \cot(\beta_{ij})) (f_j - f_i)$$

where α and β are the two angles opposite the edge (i,j) . It is symmetric, its row sums vanish (constants are in its kernel), and it is negative semi-definite — the discrete analogue of Δ . Paired with the diagonal mass matrix M of vertex areas, it is the single operator behind every section that follows. In Wolfram Language it is one sparse matrix:

```
L = DDG`cotanLaplacian[mesh];    (* V x V SparseArray *)
Max[Abs[L - Transpose[L]]]      (* 0 : symmetric *)
Max[Abs[L . ConstantArray[1., Length[L]]]] (* ~1e-15 : constants in kernel *)
```

3. Spectral geometry: hearing the shape of a mesh

The eigenpairs of the Laplacian are a mesh's natural harmonics. Solving the generalized eigenproblem

$$-\mathbf{L} \cdot \phi = \lambda \mathbf{M} \cdot \phi$$

gives eigenvalues (a shape *fingerprint*) and eigenfunctions (smooth standing waves). On the unit sphere the eigenvalues must be $l(l+1) = 0, 2, 6, 12, 20, \dots$ with multiplicity $2l+1$ — and the discrete operator delivers exactly that (2.000×3, 5.97×5, 11.8×7), while the eigenfunctions are the spherical harmonics:

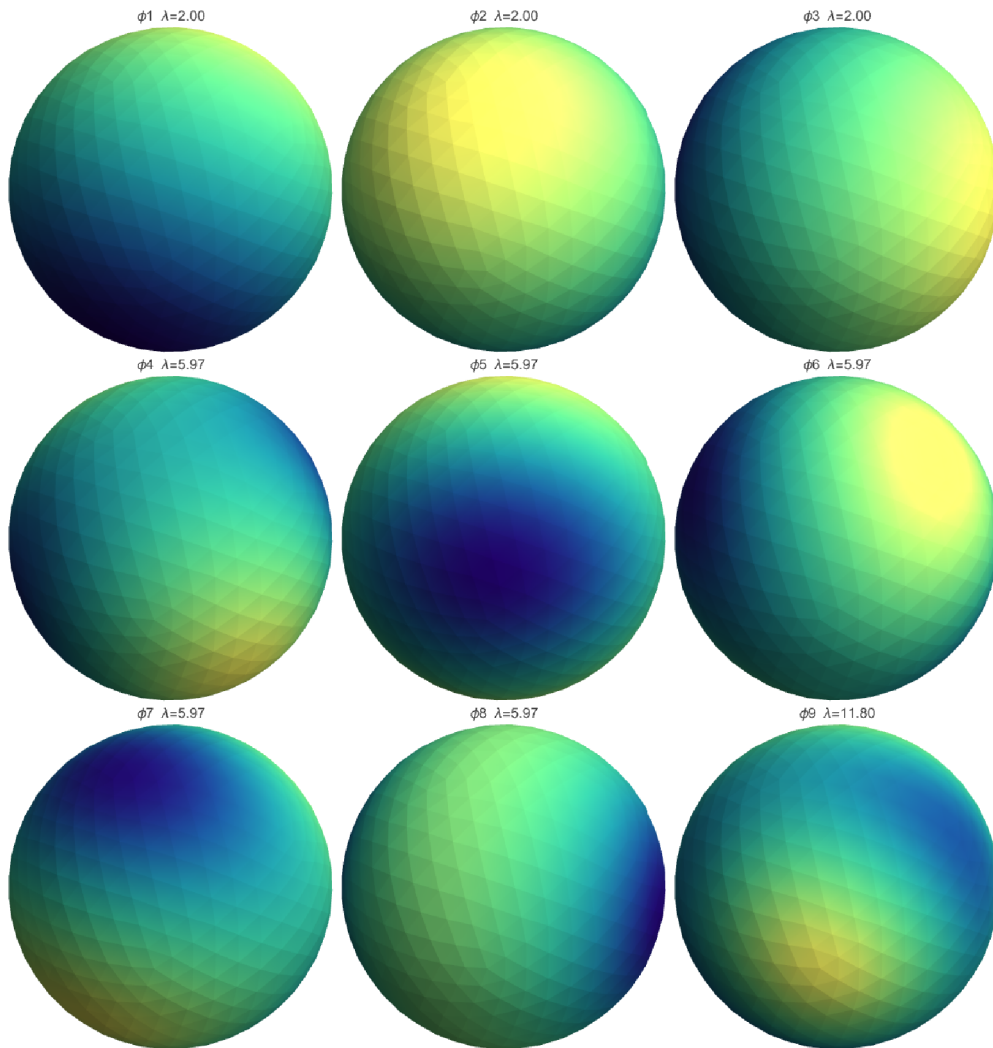


Figure 3. The first nine non-constant Laplace–Beltrami eigenfunctions on a geodesic sphere — the discrete spherical harmonics Y for $l = 1, 2, 3$.

Truncating the eigenbasis low-pass-filters geometry, exactly as a Fourier series does for signals: reconstructing a bumpy mesh from its lowest K manifold-harmonic modes recovers progressively finer detail.

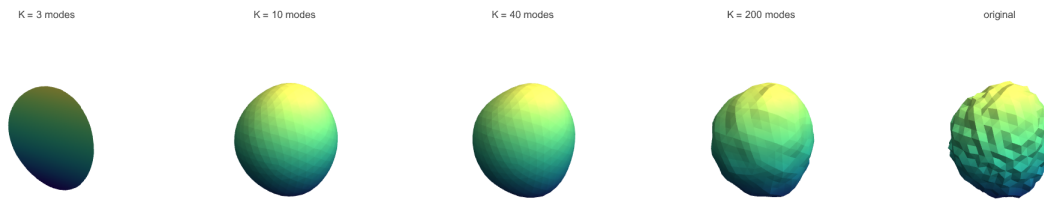


Figure 4. Spectral mesh smoothing: a bumpy sphere reconstructed from its lowest K Laplacian eigenmodes — the geometry analogue of Fourier compression.

4. Mean-curvature flow: smoothing as diffusion

Letting the Laplacian pull on the vertices themselves, $\partial x / \partial t = \Delta x$, is **mean-curvature flow**: the surface diffuses, smoothing itself by its own curvature. Explicit time-stepping is stable only for tiny steps; the semi-implicit scheme (Desbrun–Meyer–Schröder–Barr 1999) is unconditionally stable and is one linear solve per step:

$$(M - \tau L) \cdot x^{n+1} = M \cdot x^n$$

As a quantitative check, a round sphere under this flow obeys $R(t)^2 = R_0^2 - 4t$; we recover the slope -3.98 (theory -4). Applied to a bumpy sphere — and rescaled to constant volume so the eye sees *smoothing*



initial (bumpy)

after fairing (volume-preserved)

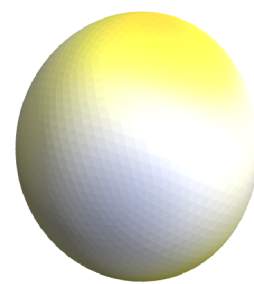
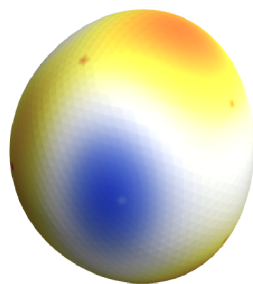
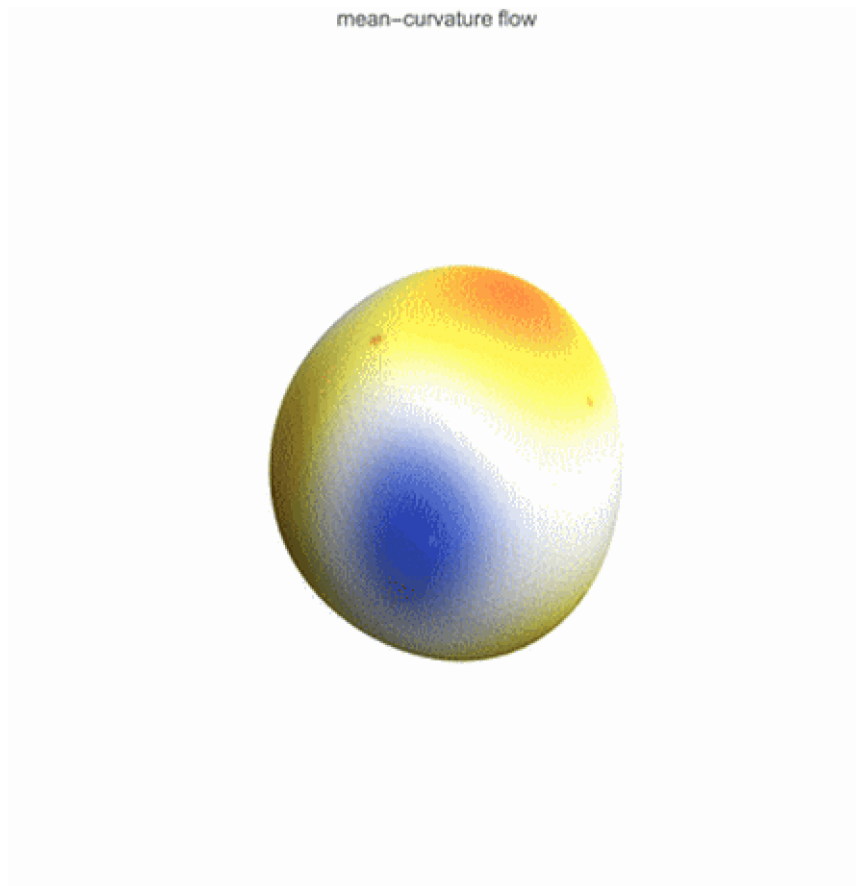


Figure 5. Mean-curvature flow (volume-preserved) faires a bumpy sphere; colour is mean curvature, which equalises as the shape rounds.



Animation. The same flow over time.

5. Outlook: geodesics and discrete exterior calculus

The same cotangent Laplacian computes **geodesic distance** via the heat method (Crane, Weischedel & Wardetzky 2013): diffuse heat briefly, normalise its gradient, then solve one Poisson problem $L\phi = \text{div } X$. And building the discrete exterior derivative and Hodge-star operators yields a **Hodge decomposition** of tangent vector fields whose harmonic part counts the surface's handles. Both are the next sections of this project.

6. References

- Meyer, M., Desbrun, M., Schröder, P., Barr, A. (2003). *Discrete differential-geometry operators for triangulated 2-manifolds*. Visualization and Mathematics III.
- Desbrun, M., Meyer, M., Schröder, P., Barr, A. (1999). *Implicit fairing of irregular meshes using diffusion and curvature flow*. SIGGRAPH '99.
- Crane, K., Weischedel, C., Wardetzky, M. (2013). *Geodesics in heat*. ACM Transactions on Graphics. cs.cmu.edu/~kmc Crane/Projects/HeatMethod
- Crane, K. *Discrete Differential Geometry: An Applied Introduction* (CMU 15-458/858 course notes). cs.cmu.edu/~kmc Crane/Projects/DDG
- Botsch, M., Kobbelt, L., Pauly, M., Alliez, P., Lévy, B. (2010). *Polygon Mesh Processing*. AK Peters.

7. Reproducibility

All code is pure Wolfram Language. The shared package `wolfram/ddg.wl` defines the mesh accessors and operators; each section's figures are produced by a standalone script. Every operator is covered by a sanity check in `tests/` (Gauss–Bonnet, mean-curvature convergence, the $l(l+1)$ spectrum, the flow rate).

```
wolframscript -file wolfram/curvature.wls      # $1 Gaussian curvature
wolframscript -file wolfram/meancurvature.wls  # $1-2 mean curvature
wolframscript -file wolfram/spectral.wls      # $3 spectrum + harmonics
wolframscript -file wolfram/meanflow.wls      # $4 mean-curvature flow
wolframscript -file community/build_notebook.wls # this notebook
```